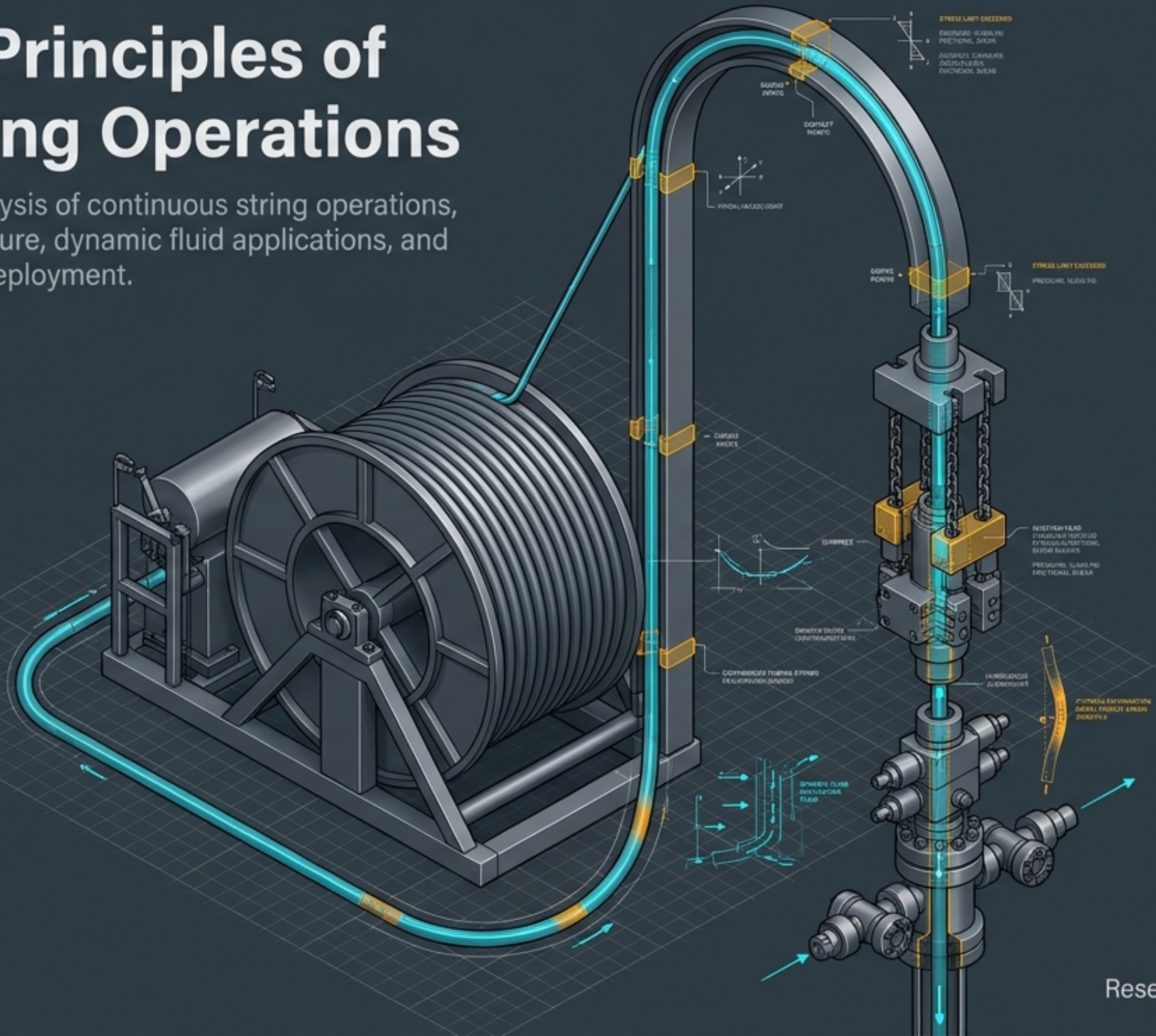


Advanced Principles of Coiled Tubing Operations

An engineering lifecycle analysis of continuous string operations, focusing on system architecture, dynamic fluid applications, and the physical boundaries of deployment.



Research Lead: Sajjad K.Hano

Continuous Interventions Eliminate the Need to Kill Production

The Paradigm Shift

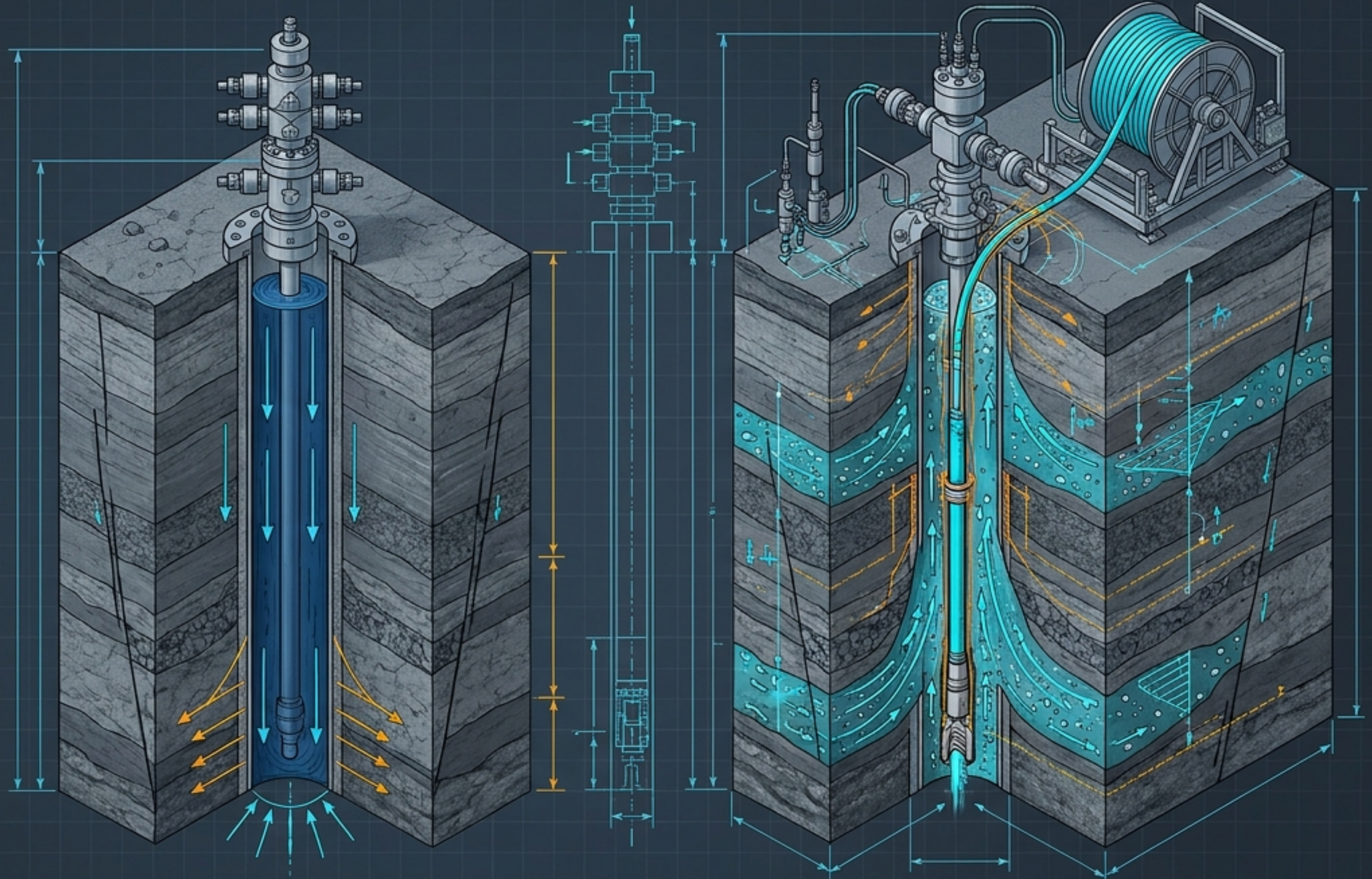
CT relies on a continuous, flexible steel pipe reeled on a large spool.

The Operational Advantage

It functions as a highly versatile intervention for well servicing, remediation, and drilling.

The Financial Impact

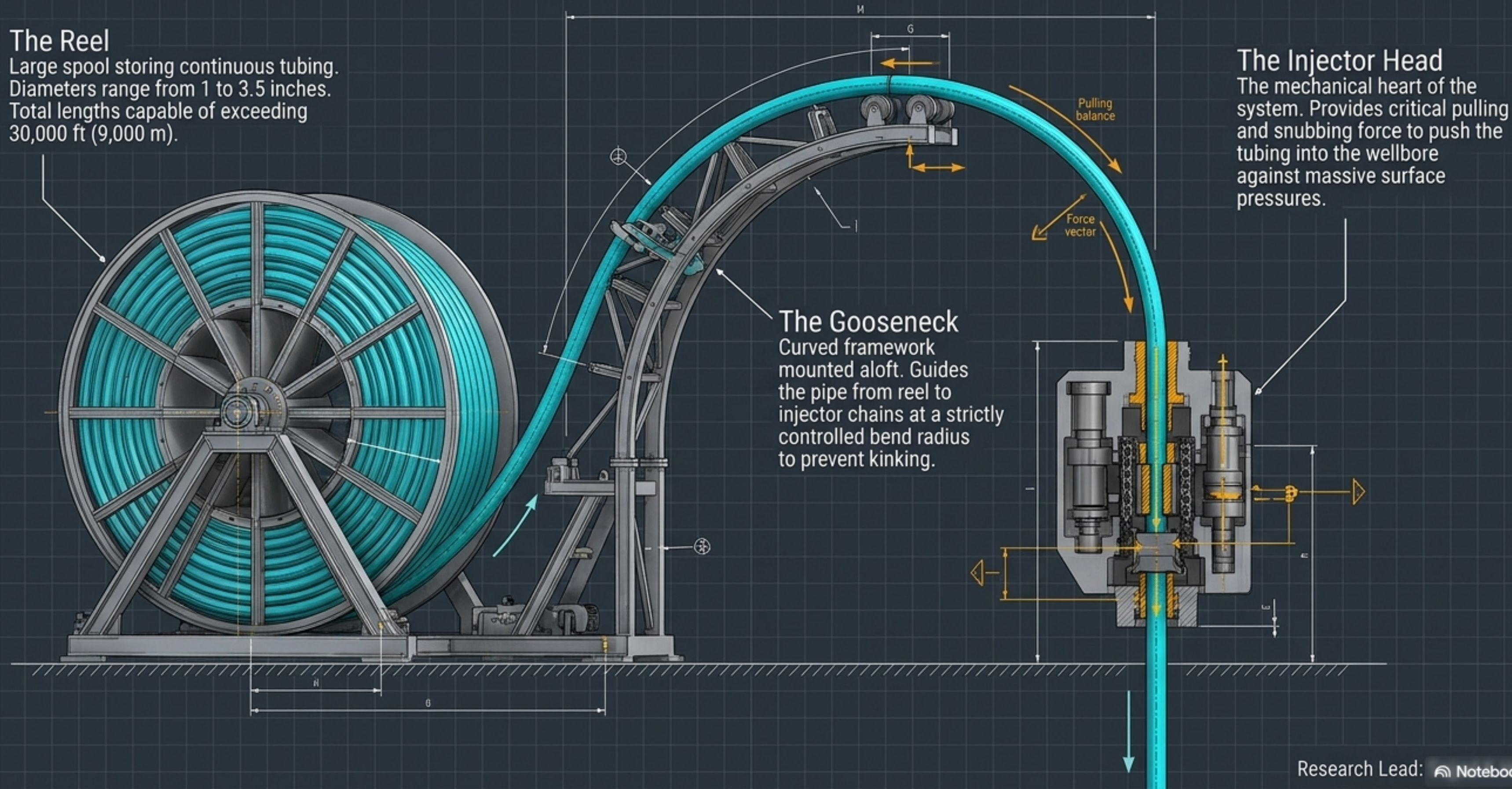
Can be run directly into a live oil or gas well without killing the well or halting hydrocarbon production.



Surface Architecture Translates Spooled Pipe to Downhole Reach

The Reel

Large spool storing continuous tubing. Diameters range from 1 to 3.5 inches. Total lengths capable of exceeding 30,000 ft (9,000 m).



Dynamic Sealing Safely Contains High-Pressure Reservoir Fluids

The Pressure Boundary

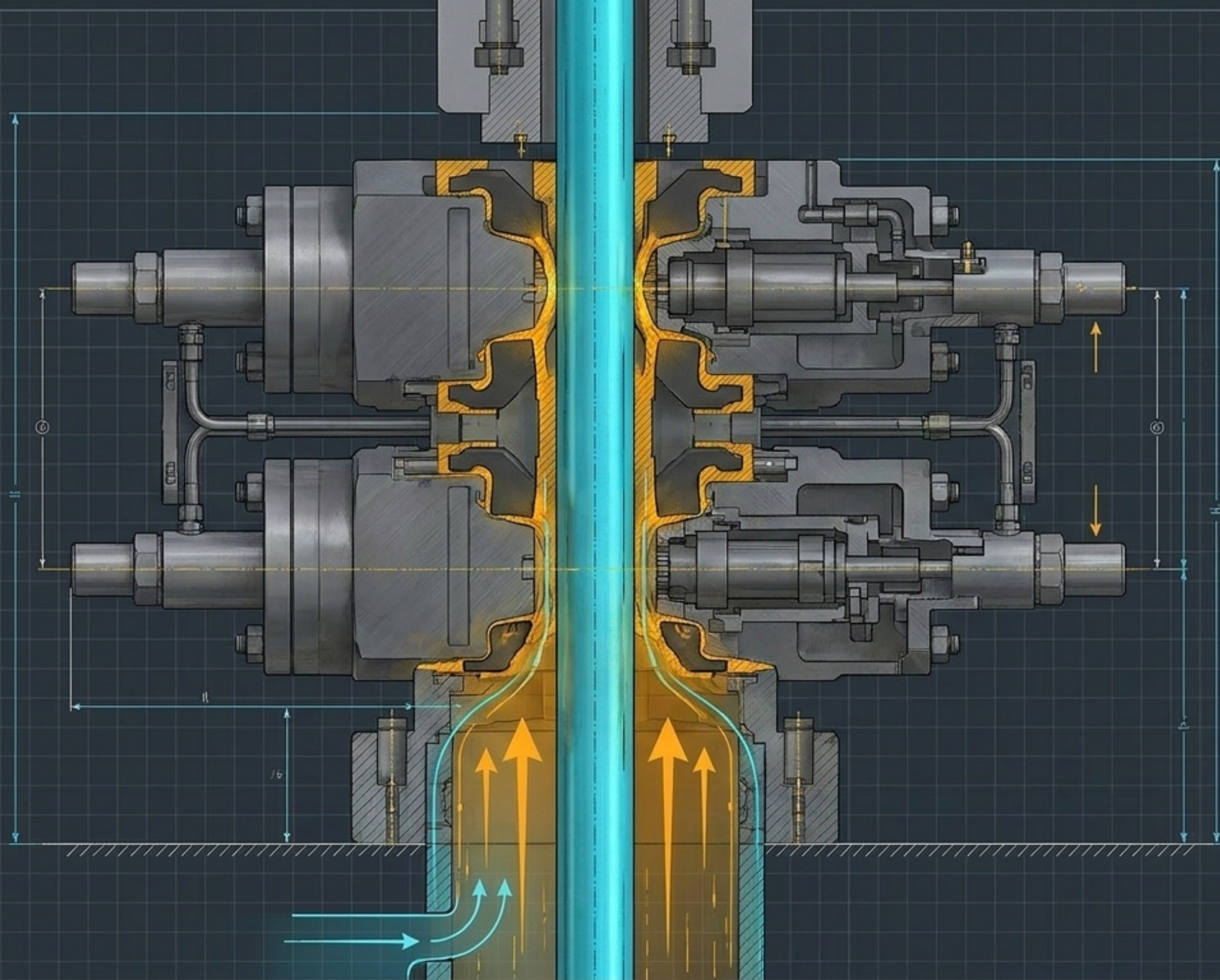
The well control stack is the ultimate safety pressure-containment system.

Dynamic Containment

Uniquely engineered to seal tightly around a moving or stationary tubing string.

Hazard Prevention

Absolutely prevents hazardous reservoir fluids and gases from escaping to the surface during deployment and retrieval.

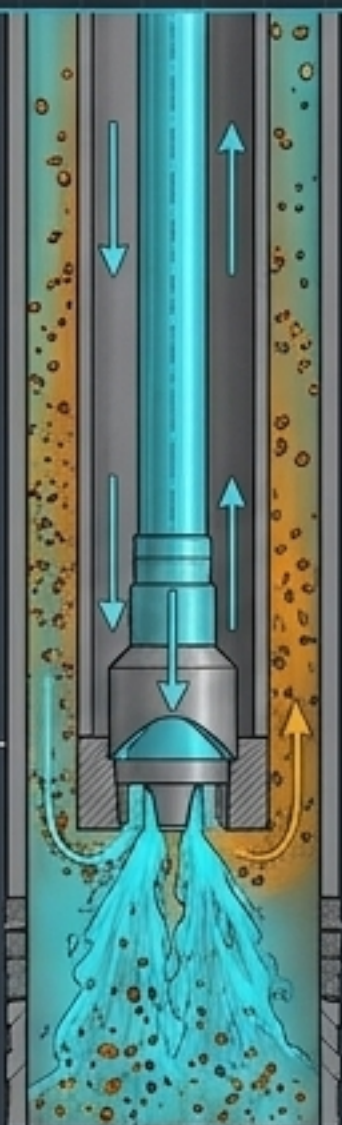


Intervention Strategies Span Hydraulic and Mechanical Domains

Stimulation

Remediation

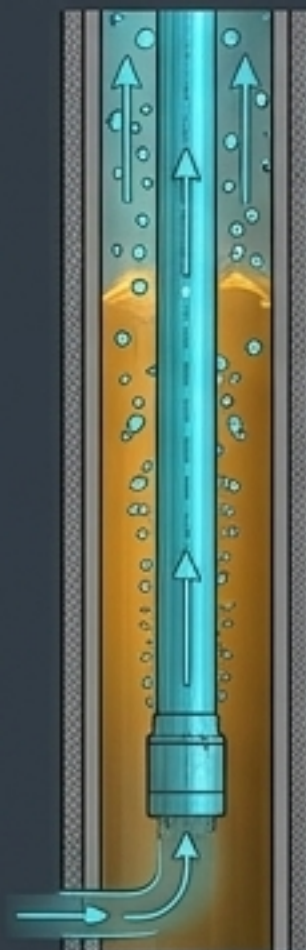
Fluid Displacement



Well Cleanouts

Pumping fluid or nitrogen to circulate accumulated sand, scale, proppant, or fill materials out of the wellbore to restore hydrocarbon flow.

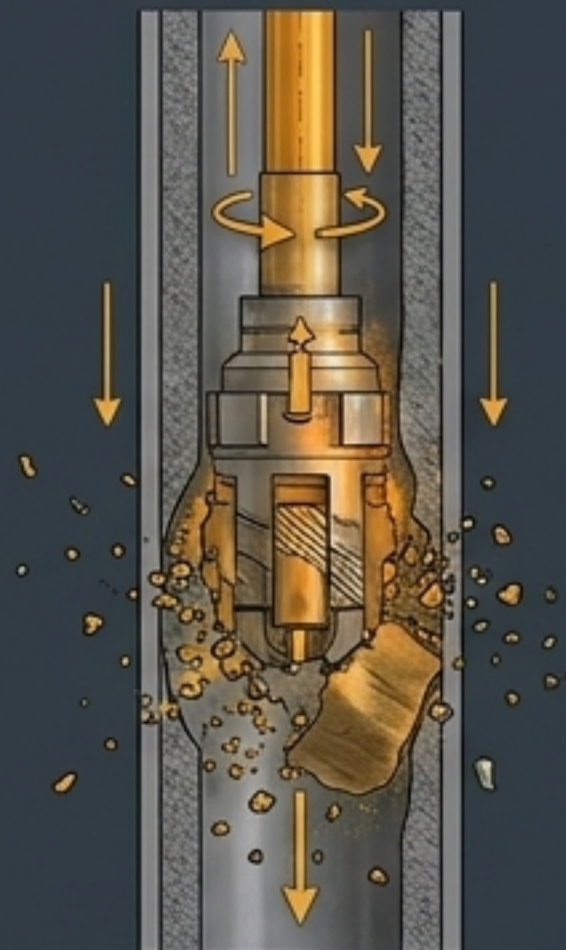
Nitrogen Lifting



Nitrogen Lifting

Deploying gaseous nitrogen to unload wells that have ceased natural flow.

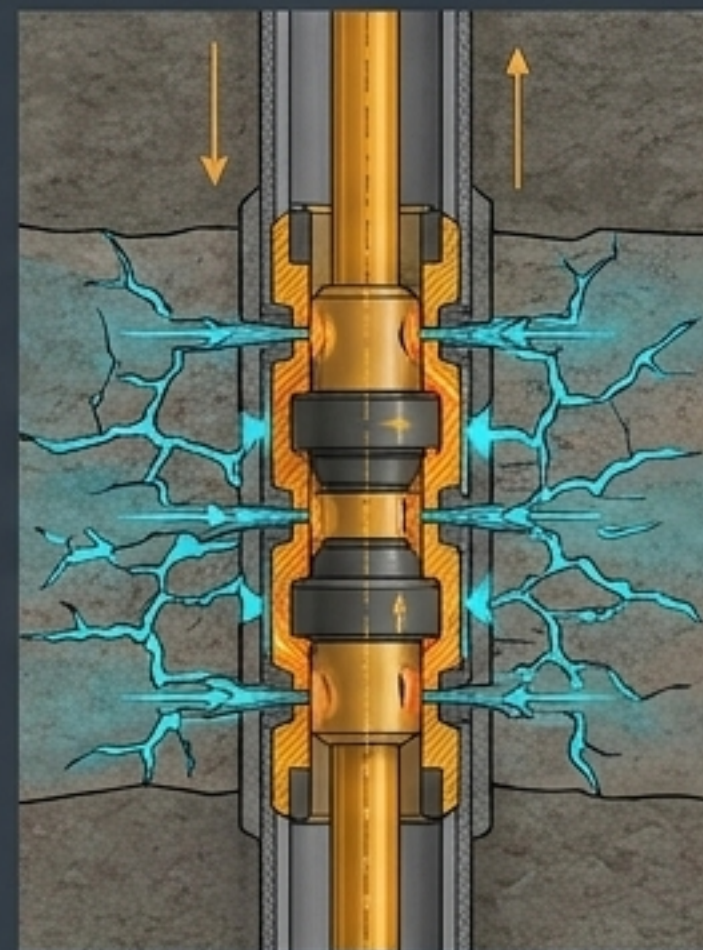
Thru-Tubing Milling



Thru-Tubing Milling

Delivering downhole tools to mechanically remove obstructions.

Matrix Acidizing



Matrix Acidizing

Pinpoint placement of corrosive treatments directly at the formation.

Nitrogen Lifts Overcome Hydrostatic Dead-Well Conditions

The Problem

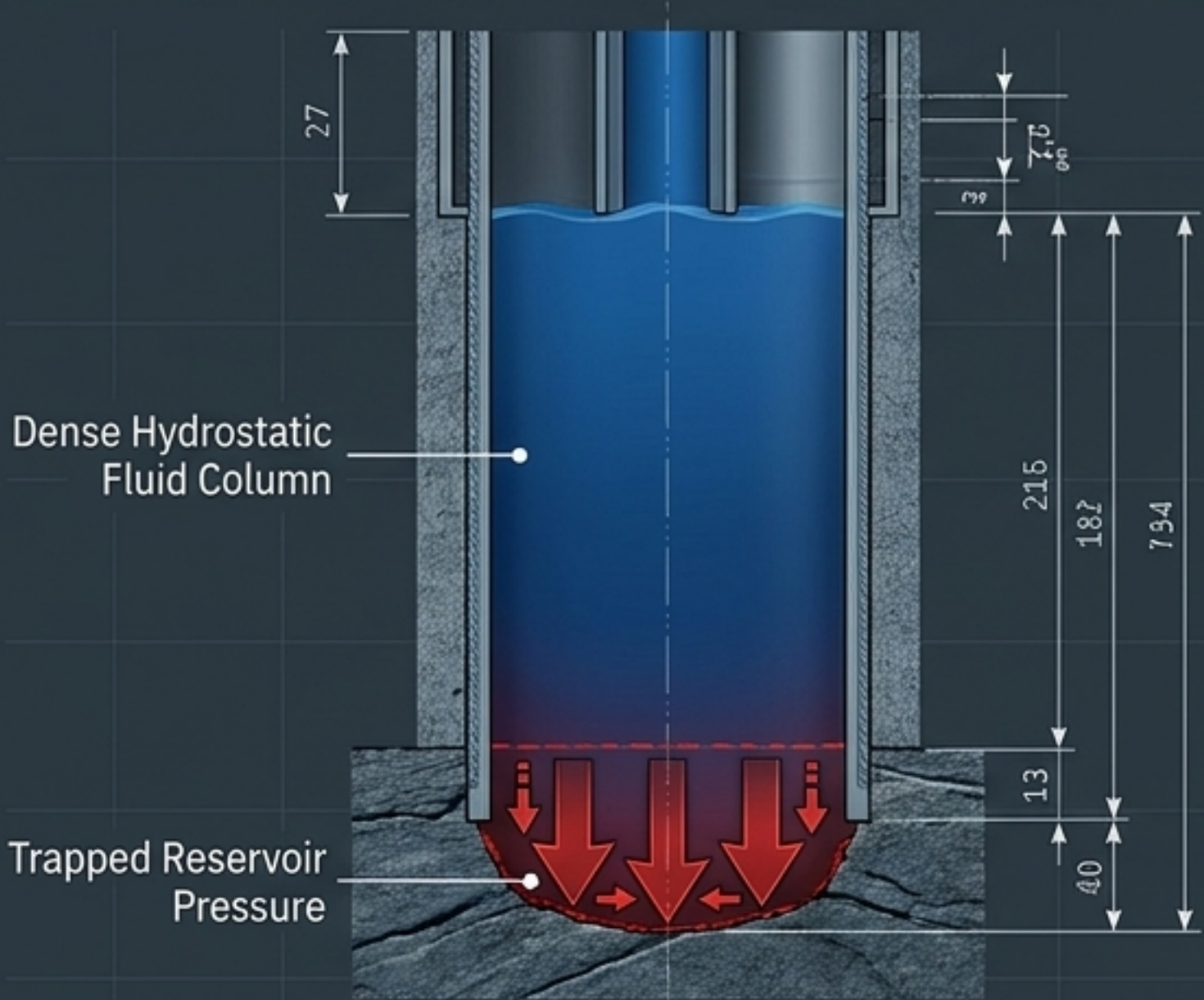
Wells cease natural flow when the hydrostatic fluid column becomes too heavy for the reservoir pressure to move.

The Catalyst

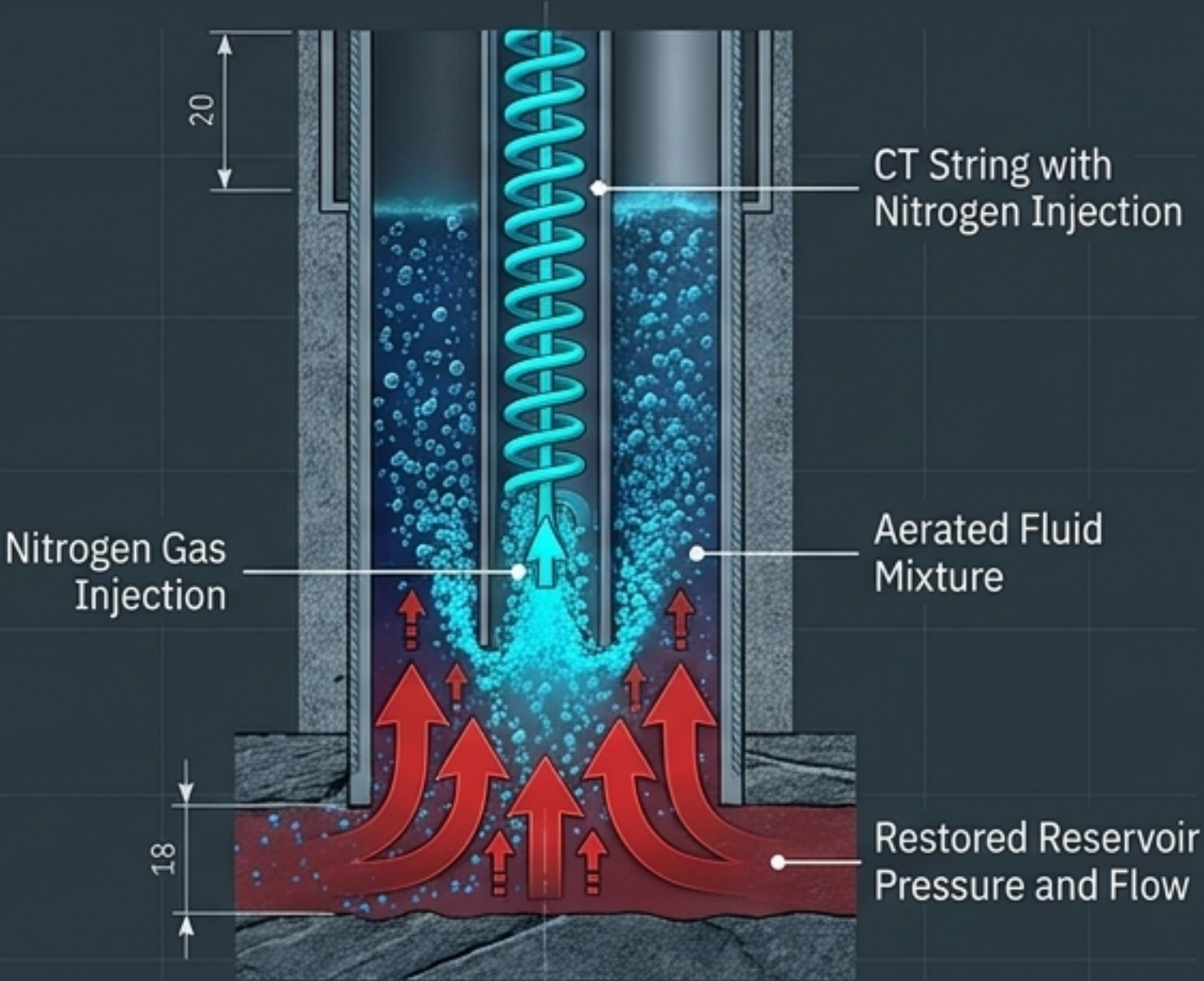
CT pumps gaseous nitrogen deep into the wellbore.

The Nitrogen Kick

This process drastically reduces the overall fluid density in the wellbore, allowing natural reservoir pressure to kick-start production.

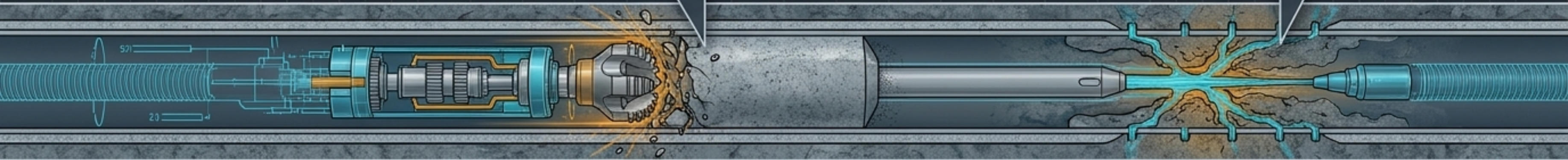
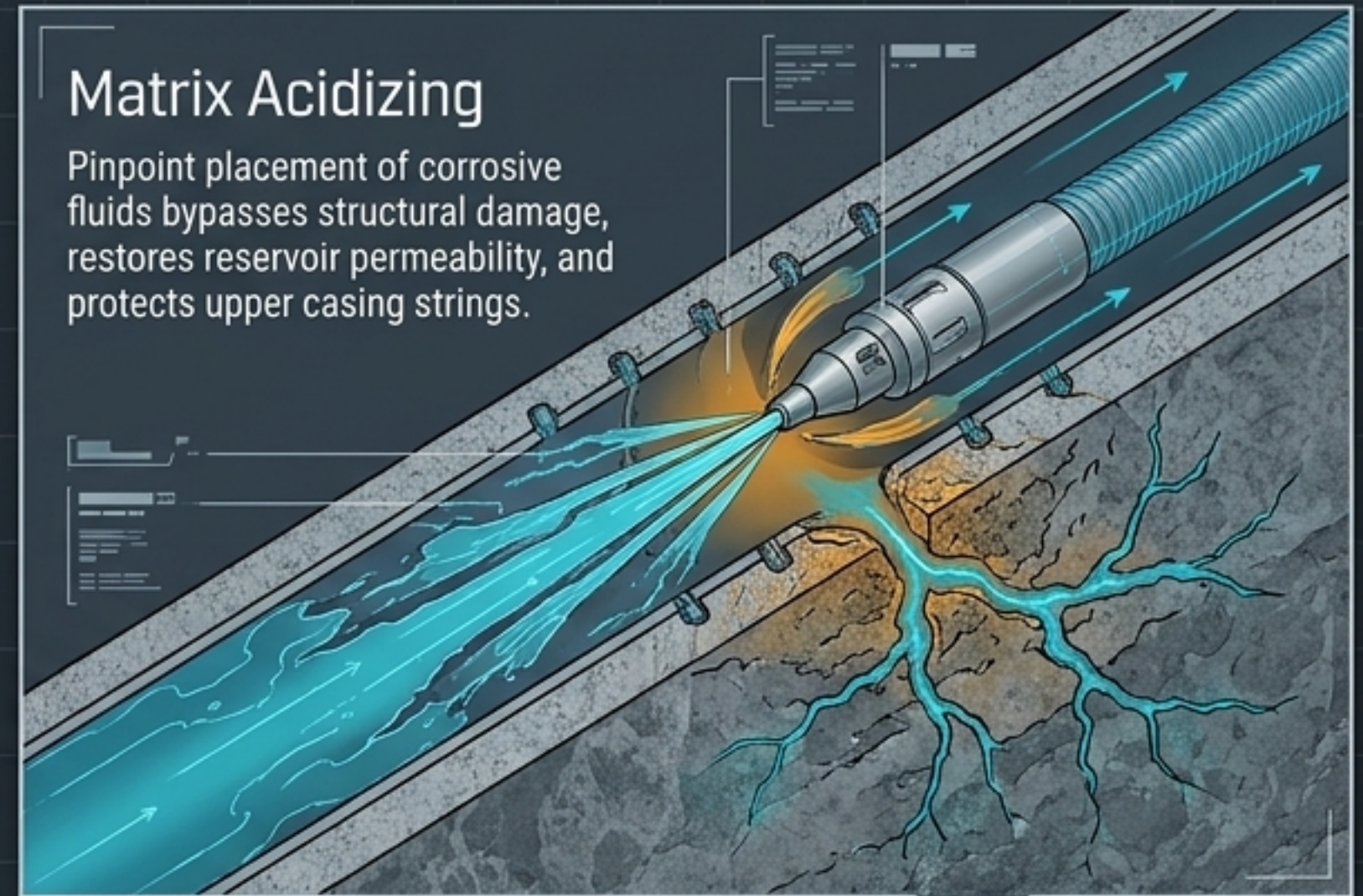
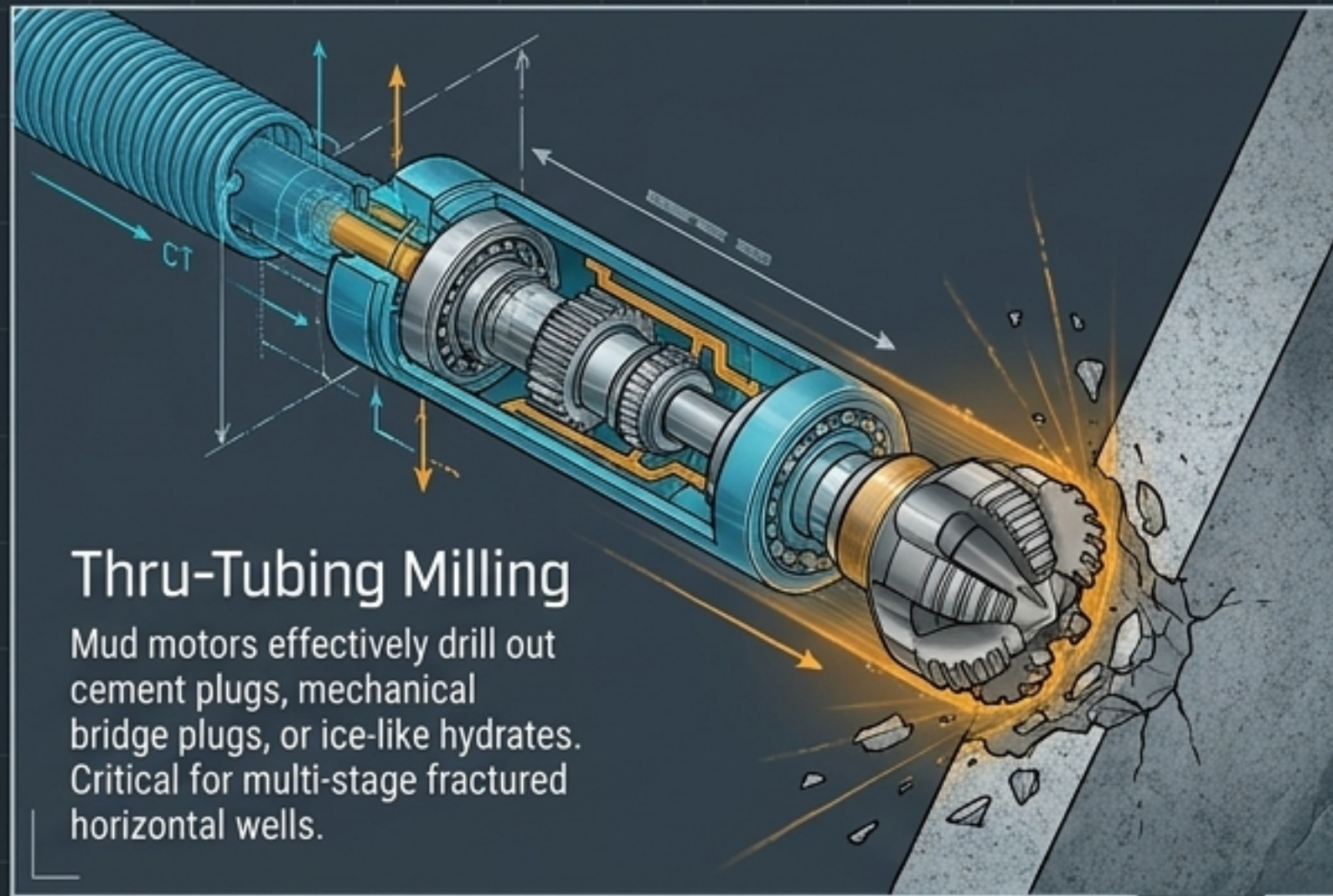


Phase 1: Hydrostatic Dead-Well Condition



Phase 2: Nitrogen Lift Activation

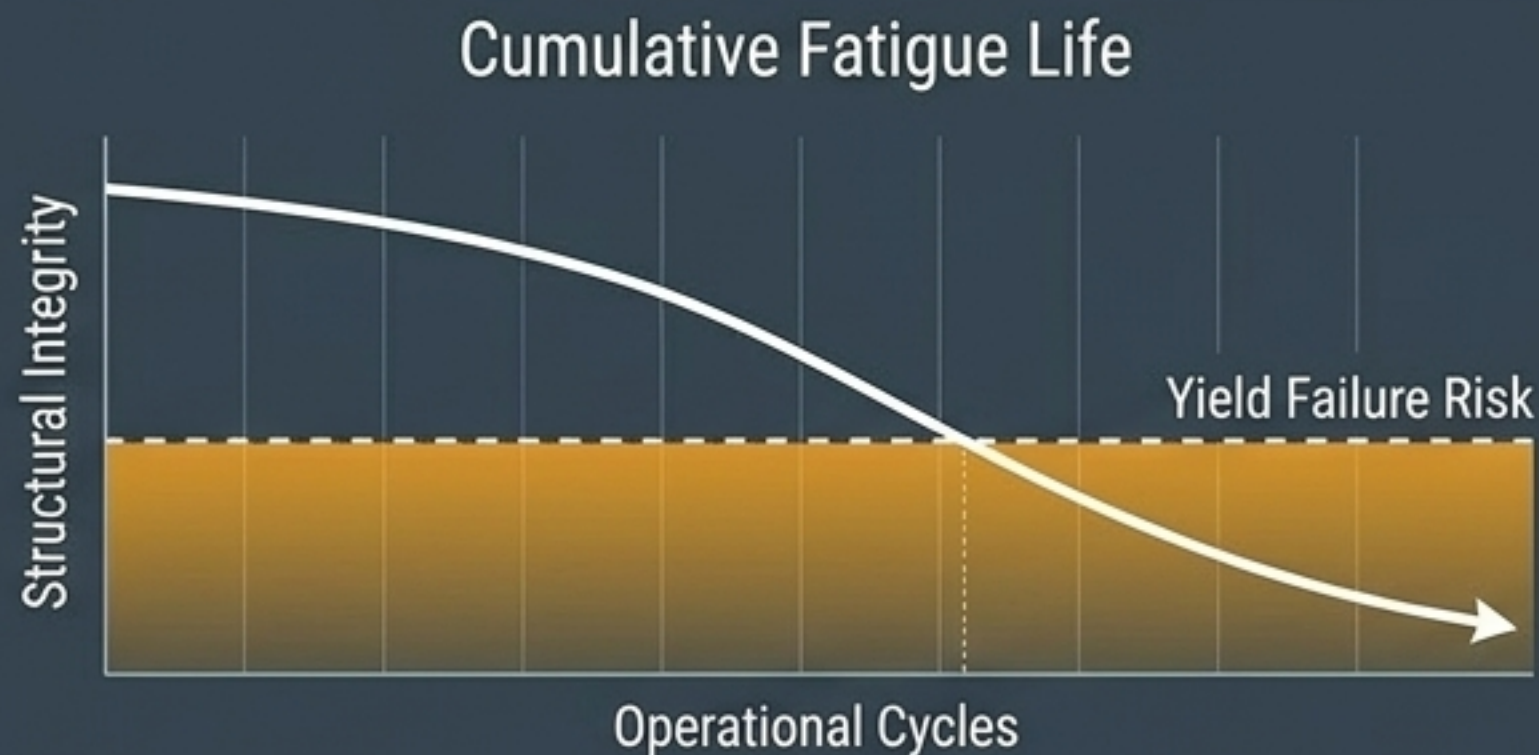
Downhole Motors and Chemical Delivery Enable Precision Remediation



Operational Parameters Dictate the Ideal Intervention Method

Feature	Coiled Tubing	Slickline	Workover Rig
Fluid Circulation Capacity	 Excellent (internal bore)	 None (solid wire)	 Excellent (jointed pipe)
Horizontal Well Reach	 Excellent (mechanically pushed)	 Poor (relies on gravity)	 Excellent (high torque/load)
Deployment Speed	 Very Rapid (continuous string)	 Rapid (lightweight)	 Slow (tripping joints)
Live Well Operations	 Yes (no kill required)	 Yes (lightweight tools)	 No (requires fluid kill)
Economic Footprint	 Moderate to High	 Low Cost	 High Rig Costs

Cyclic Plastic Deformation Defines the Functional Lifespan of the String



The Bending Cycle

The steel pipe is repeatedly bent over the reel and gooseneck, then forcibly straightened into the injector head.

Low-Cycle Fatigue

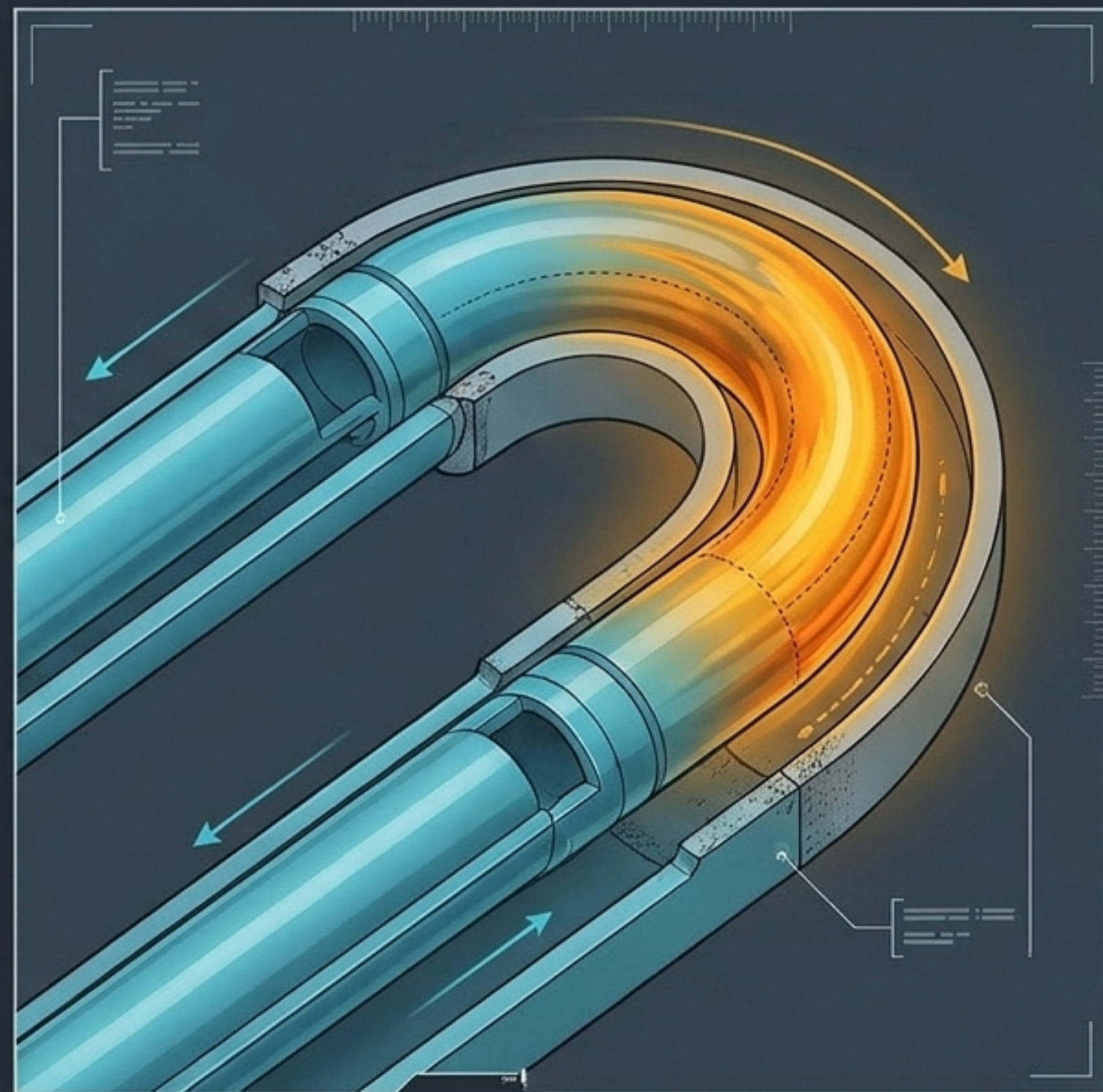
This generates massive cyclic plastic deformation with every deployment and retrieval.

Low-Cycle Fatigue

This generates massive cyclic plastic deformation with every deployment and retrieval.

Predictive Mitigation

Engineers utilize advanced predictive software models tracking the cumulative fatigue life of every single foot of the string to avoid catastrophic downhole parting.



Compressive Frictional Forces Drive the String to Ultimate Lock-Up

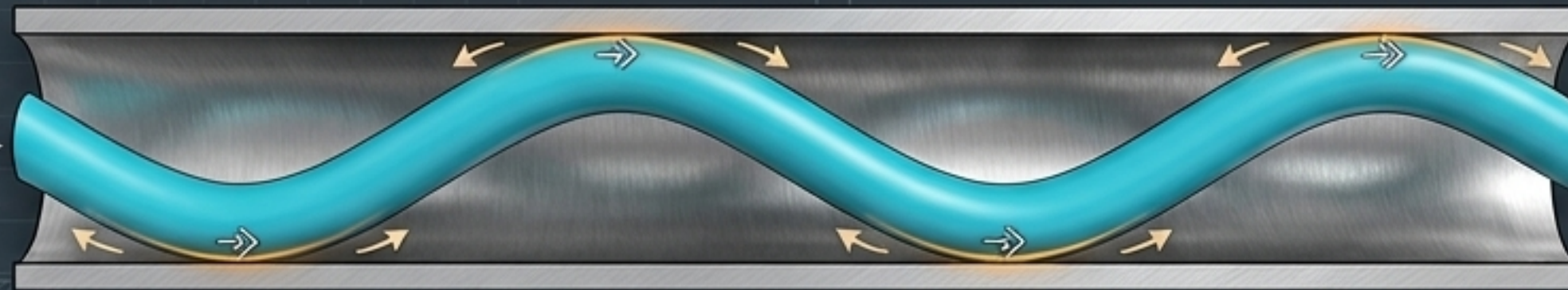
Stage 1: Frictional Drag



Compressive Force →
← Frictional Drag

Pushing CT down highly deviated or horizontal wells generates massive compressive forces.

Stage 2: Sinusoidal Buckling



Compressive Force →
← Frictional Drag

When forces exceed critical limits, the pipe warps into a 2D wave pattern.

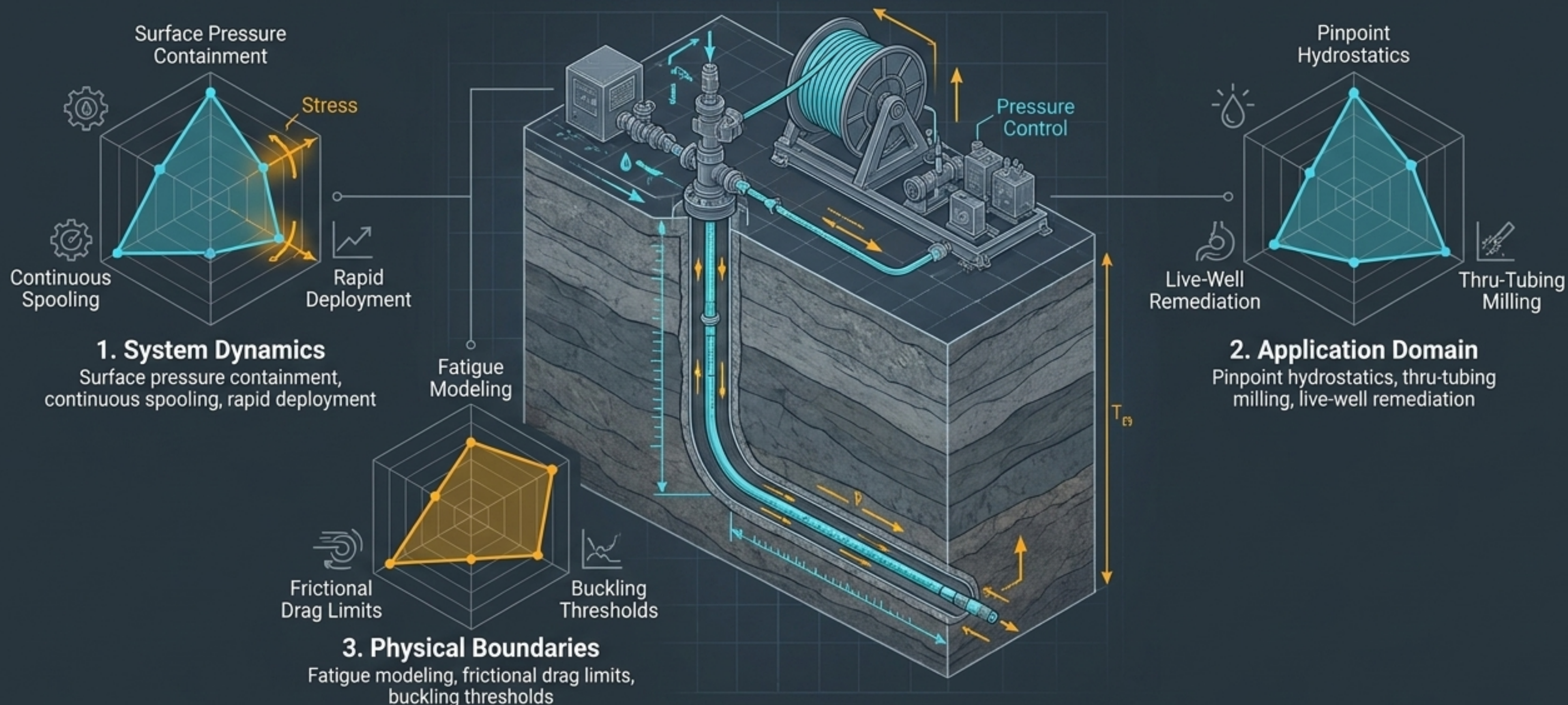
Stage 3: Helical Buckling & Lock-Up



Compressive Force →
⚠ Lock-Up Threshold
NO REACH

The pipe twists into a 3D corkscrew against the casing. Lock-up is the absolute friction threshold where no further reach is possible.

Operational Success Requires Balancing Capability Against Physical Limits



Coiled Tubing is not merely a deployment tool; it is a dynamic engineering system operating at the absolute margins of metallurgical and physical constraints. True operational mastery lies in navigating the tension between downhole reach, fluid dynamics, and cumulative material fatigue.